REQUEST FOR PROPOSAL & STATEMENT OF WORK

Predictive Maintenance IoT Platform — IndustrialOps Manufacturing

Client: IndustrialOps Manufacturing | **Vendor:** SensorAI Technologies | **Budget:** $720,000 | **Duration:** 11 months | **Start:** 2026-09-01

1. Project Overview & Goals

IndustrialOps requires an IoT-based predictive maintenance platform for 850 CNC machines across 4 factories, ingesting 500 sensor streams at 1Hz, predicting failure 72 hours in advance, and integrating with SAP PM for work order generation.

Goals: (1) Failure prediction recall ≥ 90% with 72h advance notice. (2) False alarm rate <3% to prevent unnecessary downtime. (3) Unplanned downtime reduced by 35%. (4) Platform processes all 500 streams with <5 min latency.

2. Scope of Work & MVP

MVP (Month 1–4): IoT ingestion pipeline for 50 machines (pilot factory), anomaly detection baseline, and maintenance alert API. MVP complete when system predicts 3 of 5 injected synthetic failure scenarios with 24h advance notice.

Out of Scope: Robotic process automation; autonomous work order approval; AR maintenance guidance; SCADA write-back commands.

3. Work Breakdown & Timelines

ID	Work Item	Timeline	Entry Criteria	Definition of Done / Exit Criteria
WI-001	IoT Ingestion & Edge Computing	Month 1–3	Network topology documented; edge hardware procured	500 streams ingested at 1Hz; edge pre-processing reduces ba
WI-002	Anomaly Detection & Failure Prediction	Month 2–5	WI-001 done (pilot); 18-month sensor history provided; failure labels annotated	Recall ≥ 90% at 72h; FAR <3%; predictions [illegible] confidence intervals
WI-003	SAP PM Integration	Month 4–7	WI-002 done; SAP PM RFC access; work order templates approved	Work orders auto-generated for high-confidence predictions; s
WI-004	Maintenance Dashboard & Mobile App	Month 5–8	WI-002 done; UX mockups approved	Machine health heatmap updates <5 min; mobile push alerts t
WI-005	Full-Factory Rollout (4 Plants)	Month 7–10	WI-001-004 stable in pilot; network upgraded	All 850 machines monitored; factory-specific models tuned; 35
WI-006	ICS Security Hardening	Month 9–11	WI-001-005 stable; ICS risk assessment complete	OT network segmented from IT; all ICS components patched;

4. Legal & Contractual Obligations

4.1 Late Delivery Penalty: $8,000 per calendar day for Tier-1 milestones (WI-001, WI-002, WI-005). $3,000 per day for other milestones. Vendor must provide 30-day advance notice of potential delays.

4.2 Subcontracting: Vendor may subcontract edge hardware installation and network configuration with 30-day written notice to Client. Core ML development and ICS security hardening may not be subcontracted. All subcontractors must comply with IEC 62443 requirements for industrial environments.

4.3 Indemnity: Vendor indemnifies Client for any production loss or equipment damage directly caused by erroneous maintenance alerts (false negatives resulting in missed failures). Indemnity cap: $2M per incident. Client indemnifies Vendor for losses arising from Client's failure to act on a correctly issued alert.

4.4 Intellectual Property: Vendor retains ownership of the predictive maintenance platform and base ML models. Client receives an exclusive licence for the manufacturing sector in the 4 countries of operation. Vendor may not licence the same trained models (calibrated on Client's data) to direct competitors. Client owns all sensor data and failure label annotations.

4.5 Insurance: Product Liability ≥ $10M (covering industrial automation); Professional Indemnity ≥ $5M; Cyber Liability ≥ $5M with OT/ICS coverage; Business Interruption covering factory downtime events.

5. Security Requirements

5.1 SAST: All edge firmware and cloud platform code scanned with Coverity (C/C++ firmware) and Bandit (Python cloud). Firmware builds require SAST sign-off before flashing to edge devices. Supply chain attestation (SLSA Level 2) required for all dependencies.

5.2 VAPT: VAPT before pilot go-live and full-factory rollout. Scope: cloud platform, edge devices, OT network, SAP integration, and mobile app. ICS-specific test cases (OT protocol fuzzing, Man-in-the-Middle on MQTT) mandatory.

5.3 ICS Security Assessment: IEC 62443 Zone and Conduit model must be documented before WI-006. Annual ICS penetration test by an ICS-CERT qualified firm. Any OT-scope critical finding halts production deployment until resolved and verified.

6. Project Closure

Closure: (1) Failure recall ≥ 90% sustained 60 days across all 4 plants. (2) 35% downtime reduction validated. (3) ICS security assessment complete; all criticals resolved. (4) VAPT reports clear. (5) SAP PM integration tested and accepted. (6) Maintenance team trained (40h per technician). (7) 90-day hypercare with ICS incident response SLA <2h.